PROPOSAL DEFENSE PRESENTATION GUIDE

A presentation structure for MIT / Capstone proposal defense

Reference proposal: Prescriptive Analytics-Based System for Borrowers Risk and Poverty Assessment with Scholarship Allocation for CARD, Inc.

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Recommended principle: present the argument, not the manuscript.

Presentation Principle

The defense should answer one central question:

“Why is this study needed, and why is your proposed approach appropriate?”

DO

Use concise bullets, diagrams, matrices, and process flows. Explain why each choice was made.

DON’T

Read the manuscript, overload slides with paragraphs, or present unsupported claims as final findings.

Recommended Defense Flow



The transitions matter: each section should justify the next one.

Chapter I establishes the need → Chapter II proves the gap → Chapter III shows how the gap will be addressed.

COMPLETE TITLE OF THE CAPSTONE PROJECT

Researcher/proponent
Program (MIT)
Adviser
Institution
Date

Chapter I — Background and Context

Prescriptive Analytics-Based System for Borrowers Risk and Poverty Assessment with Scholarship Allocation for CARD, Inc.

Start with the microfinance / poverty-assessment context.

Narrow to CARD, Inc. and its operational decision needs.

Introduce the need for evidence-based borrower risk assessment, PPI monitoring, and scholarship prioritization.

End with the proposed integrated decision-support system.

Core idea

Integrate predictive modeling + PPI-based poverty assessment + scholarship prioritization into one decision-support platform.

Research Problem

Main problem

CARD, Inc. lacks an evidence-based integrated system for early borrower-risk identification, PPI-based poverty monitoring, and scholarship prioritization.

Specific problem / research question
1. Predict borrower risk using available client and loan data.
2. Integrate PPI into automated poverty assessment.
3. Improve operational efficiency in client-risk and scholarship prioritization.
4. Integrate loan, PPI, and dependent datasets into a single platform.

Research Objectives

General objective

Develop an integrated system that combines poverty assessment, scholarship prioritization, and predictive modeling to accurately identify borrower risk and support informed decision-making.

01 Predictive model

Identify borrowers at high risk of default.

02 PPI mapping

Map clients using their PPI scores.

03 Vulnerability dashboard

Track changes in clients' poverty profiles.

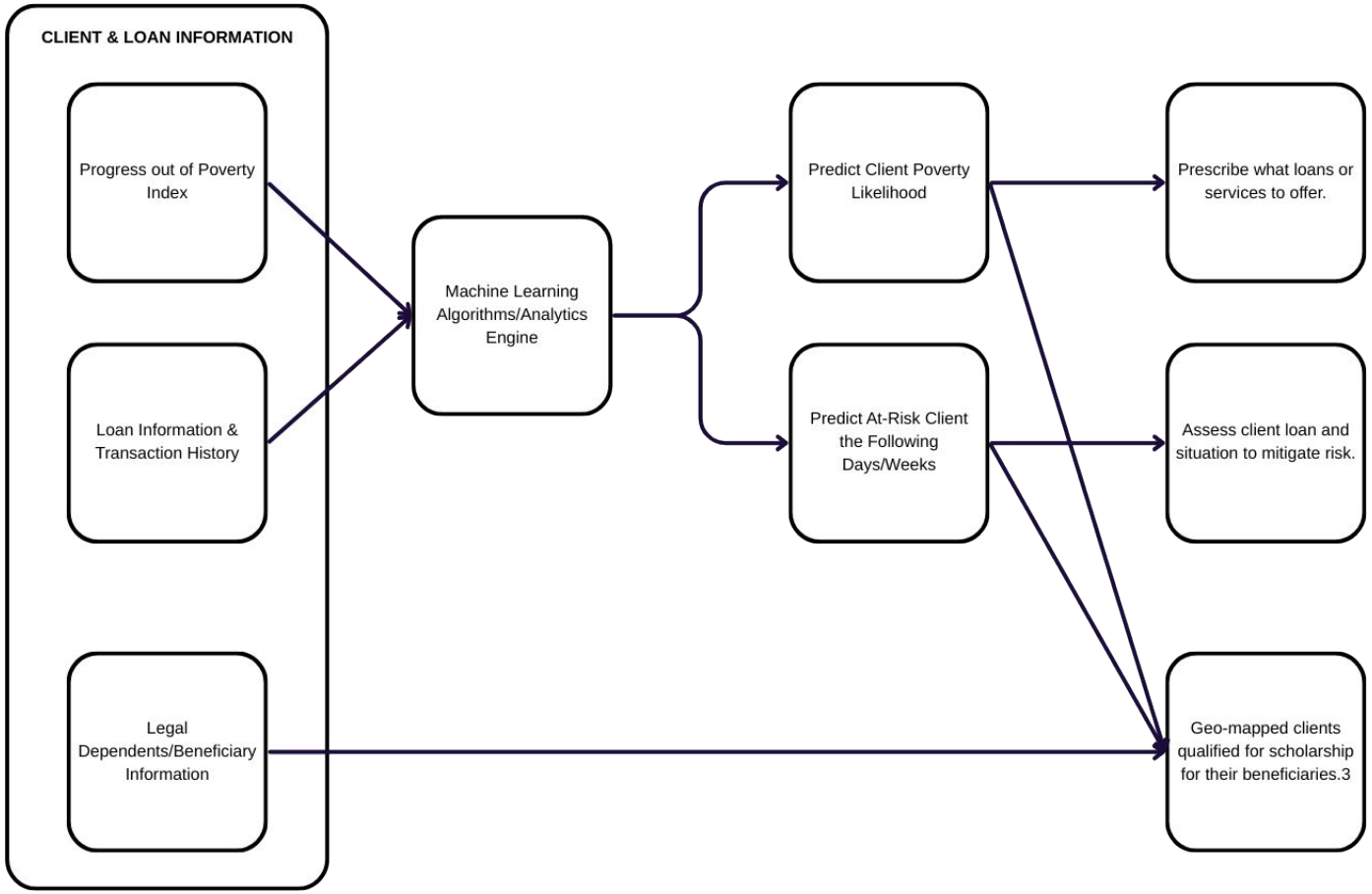
04 Scholarship prioritization

Prioritize based on PPI and risk classification.

05 Evaluation

Assess predictive accuracy, operational performance, and usability.

Research Framework



Explain the framework in layers

1) Inputs: client, loan, PPI, and dependent information.
2) Analytics: machine-learning / analytical engine.
3) Outputs: risk, poverty likelihood, and scholarship decisions.

Framework anchors

AI/ML, CRISP-DM, Data-Driven Decision Making, Decision Theory, Optimization Theory, Capability Approach, and technology-adoption perspectives are used in the manuscript.

Scope, Delimitations, and Significance

Scope

CARD, Inc. setting; client/loan information; repayment behavior; PPI results; dependent/beneficiary information; risk, poverty, vulnerability, and scholarship-prioritization functions.

Delimitations

State clearly what data, users, branches/areas, system functions, and prediction tasks are excluded or outside the study.

Significance

Emphasize practical value to CARD, Inc. decision-makers and monitoring personnel, as well as potential contribution to data-driven microfinance and social-development practice.

Chapter II — Present the Literature as a Matrix

Recommended approach

Use a thematic matrix instead of presenting long paragraphs of related literature.

Author / Year	Focus	Method / Technology	Key finding	Relevance / Gap
Balaji (2024)	Microfinance & empowerment	Microfinance program study	Improves access to finance & entrepreneurship	Supports microfinance context
Ashraf et al. (2024)	Poverty reduction	Microfinance initiative study	Positive poverty/economic effects	Justifies poverty focus
Abusharbeh (2023)	Portfolio-at-risk	Predictive modeling	Models factors related to repayment risk	Supports borrower-risk prediction
Fajardo et al. (2024)	Scholarship allocation	Machine learning	Predictive analytics can support resource allocation	Supports scholarship analytics
Alam et al. (2021)	Client segmentation	Clustering	Groups clients by behavior/demographics	Supports segmentation/analytics

Chapter II — Matrix: Predictive Analytics and Evaluation

Study	Area	Contribution to the proposal	How to present it
Oti et al. (2023)	Clustering	K-Means / hierarchical clustering reveal financial patterns	Supports client segmentation
Almeida et al. (2023)	Forecasting	Time-series analysis for financial performance	Supports trend monitoring
Li & Law (2023)	Forecasting	Forecasting supports financial sustainability / repayment monitoring	Supports longitudinal analysis
Shubho et al. (2022)	Visualization	Real-time visualization improves monitoring	Supports dashboards
Vujović (2021)	Model evaluation	Accuracy, precision, recall, specificity, AUC	Supports evaluation plan
Naidu et al. (2023)	Financial ML evaluation	Metrics are important in risk-related applications	Supports model reliability

Presenter emphasis: literature should justify the METHOD, not merely prove that the topic is popular.

Chapter II — Matrix: The Integration Gap

Study	Risk	Poverty	Scholarship	Integrated system?
Dako et al. (2024)	✓	Related	—	No
Illangakoon (2024)	✓	Related	—	No
Iligan et al. (2023)	—	Related	✓	No
Gutiérrez-Nieto & Serrano-Cinca (2023)	Related	✓	—	No
Anuar et al. (2024)	Related	✓	—	No
Present study	✓	✓	✓	YES

GAP **The reviewed literature treats borrower risk, poverty assessment, and scholarship support largely as separate or related interventions rather than one integrated analytics-driven system.**

Chapter II — Synthesis and Research Gap

What the literature establishes

Microfinance supports poverty reduction, financial inclusion, and social development.

What technology adds

Predictive analytics, machine learning, clustering, forecasting, and visualization can strengthen monitoring and decision-making.

What remains missing

An integrated system connecting borrower risk, poverty assessment, and scholarship prioritization.

Therefore…

The proposed study addresses the gap by integrating predictive modeling, PPI-based assessment, and data-driven scholarship prioritization into a single decision-support platform for CARD, Inc.

Chapter III — Research Design and Locale

Research design

Experimental research design is used to develop and compare predictive models for borrower risk and PPI estimation, with evaluation under controlled conditions.

Locale

CARD, Inc. (A Microfinance NGO), San Pablo City, Laguna, Philippines. The study uses operational data relevant to client behavior and vulnerability.

Why this fits

The design allows candidate algorithms to be compared using predefined performance metrics while the setting provides real operational context.

Applied Concepts and Algorithm Analysis

Component	Purpose	Candidate / approach	Evaluation / output
Borrower risk	Identify at-risk borrowers	Logistic Regression, Random Forest, Decision Tree, Gradient Boosting	Accuracy, Precision, Recall, F1, AUC
PPI assessment	Estimate / classify poverty condition	Predictive modeling using selected client/PPI indicators	Defined predictive metrics
Client analytics	Identify patterns / segments	Clustering / analytical methods where applicable	Segment quality / interpretation
Scholarship prioritization	Turn information into action	Rule-based prioritization using PPI, risk, dependents, education	Rule/test-case validation

Critical defense point: do not imply that every algorithm performs every task.

Data Collection and PPI

PROGRESS OUT OF POVERTY INDEX (PPI) scorecard for the Philippines is based on the 2004 Annual Poverty Indicators Survey (APIS) V.6142011

Indicator	Pakibilugan ang mga sagot ng kinapanayam/interviewee (Response)						Score
1. Ilan po sa inyong pamilya ang nasa **edad 14 pababa?**	Lima o higit pa	Apat	Tatlo	Dalawa	Isa	Wala	
	0	4	9	15	20	26	
2. Lahat po ba ng batang may **edad na 6-14** sa inyong pamilya ay **pumasok sa eskwelahan?**	Hindi		Oo		Walang bata na may edad na 6-14		
	0		2		4		
3. Ano po ang **pinakamataas na antas** ng pag-aaral ang **natapos ng babaeng puno ng pamilya/asawang babae?**	Nagtapos sa elementarya o mas mababa pa	Una hanggang ika-apat na taon sa hayskul	Nagtapos sa hayskul	Unang taon sa kolehiyo o higit pa, o walang babaeng puno ng pamilya			
	0	3	6	11			
4. Mayroon po bang miyembro ng inyong pamilya ang tumatanggap ng **regular na sweldo?**	Wala		Meron				
	0		5				
5.Anong materyales po gawa ang inyong **dingding?**	Magaan na materyales (cogon, nipa o sawali, kawayan, anahaw)		Matibay na materyales (bakal, *aluminum*, tile, konkreto, tisa, bato, kahoy, asbestos)				
	0		4				
6. Anong materyales po gawa ang inyong **bubong?**	Magaan na materyales (gamit na makeshift, cogon, nipa, o anahaw)		Matibay na materyales (yero, *aluminum*, tile, konkreto, tisa, bato o asbestos)				
	0		2				
7.Ano po ang **uri ng palikuran o toilet** mayroon ang inyong pamilya?	Wala, open pit, close pit, o iba pa		De buhos (water sealed)				
	0		7				
8. Mayroon po ba kayong **gumaganang refrigirator? Sa inyo po ba ito?**	Wala		Meron				
	0		10				
9. **Ilan** po ang **pag-aari ninyong telebisyon? Gumagana** po ba ang mga ito?	Wala		Isa		Dalawa o higit pa		
	0		6		21		
10. Mayroon po ba kayong **gumaganang washing machine? Sa inyo po ba ito?**	Wala		Meron				
	0		10				
						TOTAL POVERTY SCORE	

Major data inputs

- Client information
- Loan and repayment records
- PPI results / indicators
- Dependent / beneficiary information

Presentation point

Explain where the data come from, how they are prepared, how privacy is protected, and how the variables support the objectives.

Data Model Generation

Insert the actual ERD / data model here.

Identify the core entities.

Show relationships among client, loan, repayment, PPI, dependent, and scholarship-related data.

Explain which fields become model features or decision variables.

Explain how the structure supports batch processing, dashboards, and reporting.

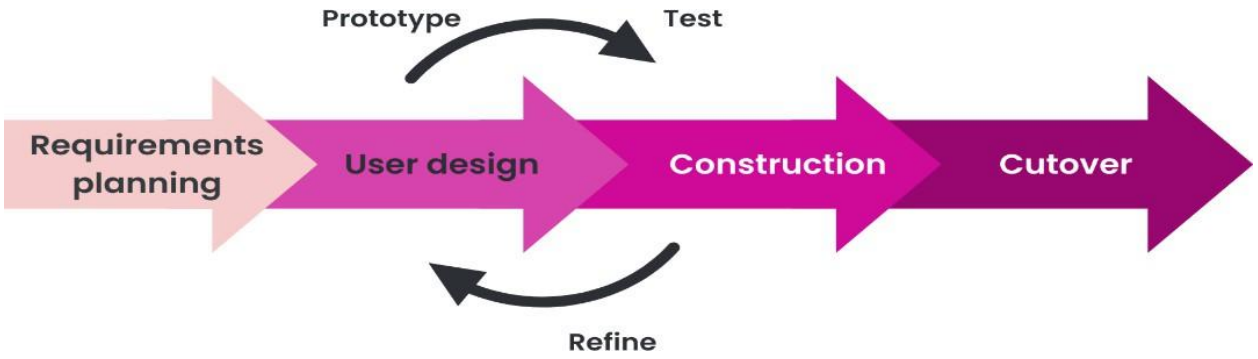
Rule of thumb

Every major objective should have the data needed to produce its intended output.

Defense question to prepare

“Why are these variables/entities necessary, and how do they relate to the prediction or prioritization task?”

System Development Methodology and Tools



Technology stack

Frontend: HTML, CSS, JavaScript, Bootstrap
Backend: Python-based framework / REST API
Database: PostgreSQL
Analytics: machine-learning and analytical engine

Rapid Application Development (RAD)

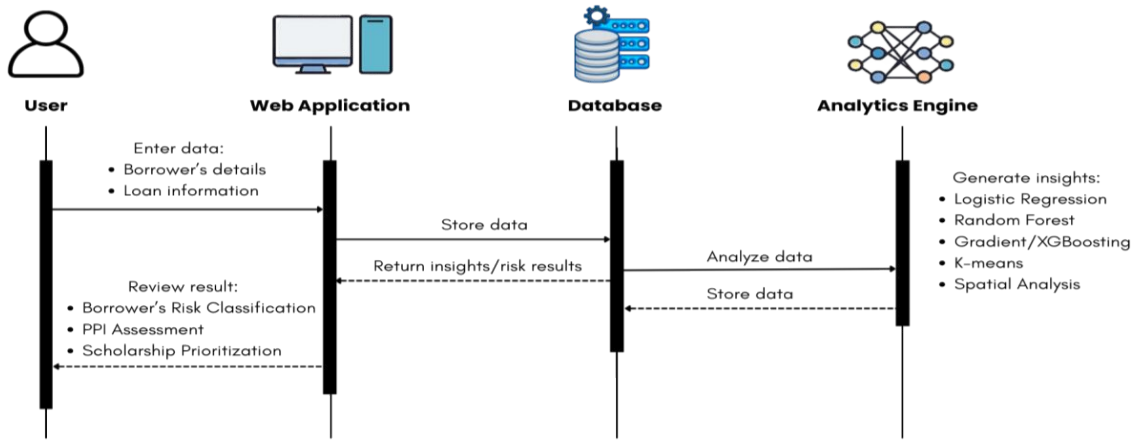
Requirements planning → User design → Construction → Cutover, with prototyping/testing and refinement.

Explain why

The tools should be justified by system requirements: interface, API/data flow, storage, analytics, and deployment.

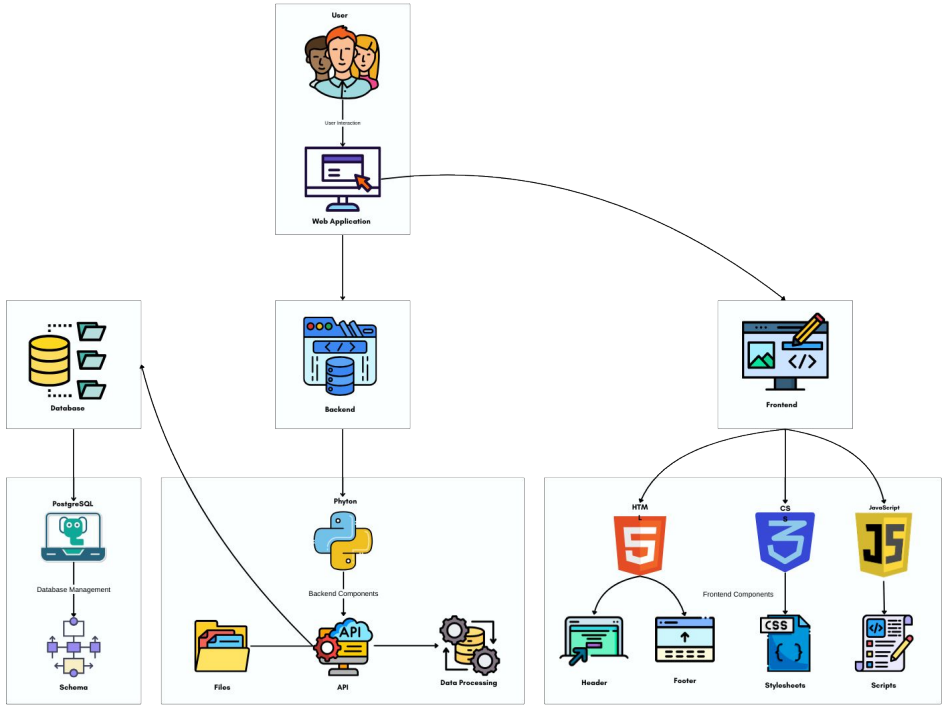
System Design

FIGURE 4 — SEQUENCE



Shows the ORDER OF INTERACTIONS: user → application → database → analytics → results.

FIGURE 5 — SYSTEM ARCHITECTURE



Shows the STRUCTURE OF THE SYSTEM: components, layers, technologies, and communication paths.

Software Testing and Evaluation

Evaluation	What it checks	Presentation emphasis
Unit Testing	Individual modules / functions	Correctness before integration
Integration Testing	Interaction among major modules	Data flow, storage, analytics, dashboards, reports
Performance Testing	Responsiveness and stability	Response time / processing behavior
User Acceptance Testing	Actual user acceptance and operational fit	Task completion, usability, acceptance
Usability / Acceptance Survey	PSSUQ + TAM-based assessment	System usefulness, information/interface quality, perceived usefulness/ease of use

PRESENT YOUR OWN TEST CASES OR EVALUATON TOOL

Proposal Defense Readiness Checklist (NOT INCLUDED IN YOUR PPT SLIDE)

- Title, problem, objectives, and proposed system are aligned.
- Every specific objective has a corresponding method or system feature.
- Chapter II uses a focused literature matrix and ends with a clear gap.
- Data sources, variables, access, privacy, and preparation are defensible.
- Each algorithm has a defined purpose and evaluation metric.
- Architecture, sequence diagram, database, API, and analytics components are consistent.
- Testing and evaluation procedures are measurable and feasible.
- Presenter can explain every figure without reading the slide.
- Claims are framed as proposed methods/expected contributions—not final findings.

FINAL MESSAGE

“The proposal is a logical response to an identified gap, supported by literature and implemented through a feasible methodology.”